



Riemannian and Complex Geometry

Lecturers: Daniel Galviz & Bowen Liu

Teaching Assistants: Anis Bousclet & Julian Alzate

PhysicsLatam.com/RC

Submission deadline: 26 November 2025, 5:59 pm (GMT-4)

— Exercises —

5.1 Complex manifold

Exercise 5.1. Show that any oriented Riemannian surface admits a natural almost complex structure.

Exercise 5.2. Show that $\mathbb{C}^n$ does not have any compact submanifolds of positive dimension. (This is in contrast to the real situation, where any manifold, compact or not, can be realized as a submanifold of some $\mathbb{R}^N$ by Whitney embedding theorem.)

Exercise 5.3. Show that any holomorphic map from $\mathbb{CP}^1$ into a complex torus is constant. What about maps from $\mathbb{CP}^n$ into a complex torus?

Exercise 5.4. Check that Plücker embedding is injective. As a consequence, Grassmannian manifold is projective.

5.2 Holomorphic tangent bundle

Exercise 5.5. For $\mathbb{C}^n \cong \mathbb{R}^{2n}$ with coordinate $z^i = x^i + \sqrt{-1}y^i$, one has

$$dx^1 \wedge dy^1 \wedge \cdots \wedge dx^n \wedge dy^n = \left(\frac{\sqrt{-1}}{2}\right)^n dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n.$$

Exercise 5.6. Let X be a complex manifold. Prove that $(T_{\mathbb{R}}X, J)$ is isomorphic to $T^{1,0}X$ as a complex vector bundle.

Exercise 5.7. Are there holomorphic vector fields on $\mathbb{CP}^n$ which vanish only in a finite number of points? If yes, in how many?

Exercise 5.8. Prove the Euler sequence on $\mathbb{CP}^n$, that is, an exact sequence of holomorphic vector bundles

$$0 \rightarrow \mathcal{O}_{\mathbb{CP}^n} \rightarrow \bigoplus_{i=0}^n \mathcal{O}_{\mathbb{CP}^n}(1) \rightarrow T\mathbb{CP}^n \rightarrow 0.$$